AGENTIC AI LEADS

THE COMPLETE APP GUIDE

Signal-based lead generation for tech recruitment agencies. Pay per showed meeting. This guide documents every engine, every screen, every rule, and every integration of the operating system behind the business — with a complete worked example.

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How to read this: if you're new, read Parts 1–2 and the Worked Example (Part 8), then use the rest as reference. The in-app Help page mirrors Part 3 and is always up to date with the code.

// PART 1

The Big Picture

The business: Agentic AI Leads books qualified sales meetings for small tech recruitment/staffing agencies (2–20 people, US/UK/FR). Clients pay only for meetings that actually happen: first meeting free, then $150 per showed meeting. No retainer, no contract.

The insight the whole machine is built on: when a company opens a new job posting, that is a public, verifiable, time-stamped signal that they need to hire — which means they are, right now, a warm prospect for a recruitment agency. Messages that open with such a signal ("saw you just opened a Senior Java role") get 15–25% reply rates versus ~3% for generic cold outreach.

The two pipelines — never mixed

The app runs two completely separate pipelines. Keeping them separate is a hard rule: they share no files, no stages, no status keys.

	LEADS pipeline	PROSPECTS pipeline
Who it finds	Companies actively hiring (from their job boards)	Recruitment agencies we want as OUR clients
For whom	Our clients (recruitment agencies)	Ourselves
Source files	watchlist.json → signals.csv	prospects.csv, agency_watchlist.json → prospect_signals.csv
Worked on	Outreach Queue + Daily Inbox pages	Prospects page
Stages	new → contacted → replied → won	new → contacted → replied → call_booked → client_won
Outcome	A booked meeting we bill $150 for	A new client paying us per meeting

How data flows (leads pipeline)

	Stage	What happens
1	WATCHLIST	watchlist.json holds job boards on 4 ATS platforms: Greenhouse, Lever, Ashby, Workable. Grown by the Discovery engine, bulk-paste, or URL paste.
2	NIGHTLY SWEEP	signal_engine.py fetches every board's public API and diffs against the last snapshot (signals_state.json). First sight of a board = silent baseline.
3	SIGNAL + HOOK	Each newly opened role becomes a dated signal with a ready-made opener sentence, appended to signals.csv.
4	PER-CLIENT FILTER	filter_engine.py passes every signal through every active client profile: exclusions, inclusions, locations, then scoring.
5	ENRICHMENT	enrichment_engine.py (Apollo) or the manual contact form names the decision-maker: name, title, LinkedIn. Cached per company forever.
6	OUTREACH QUEUE	Follow-ups due first, then ranked new sends. You copy the drafted DM, send it on LinkedIn, click Sent. The app tracks all timers.
7	MEETING = MONEY	Replies handled by a human within 2 hours. A showed meeting = stage won = billable at the client's meeting rate.

The PROSPECTS pipeline mirrors this flow with agency boards: an agency posting a new role means *they just signed a client mandate* — which is exactly the moment to offer them more clients.

Stage colors (used identically everywhere)

new	contacted	replied	won	dismissed

// PART 2

Quick Start

One-time setup

```
pip install -r requirements.txt
cp .env.example .env # paste APOLLO_API_KEY when you have one
python signal_engine.py --baseline # first ever run: records state silently
streamlit run app.py # opens the CRM at localhost:8501
```

Then create at least one client profile in **System** → **Client Configs** (your own prospecting profile counts), and make sure the Watchlist isn't empty. The Home page's **setup checklist** verifies all four requirements and names the fix for any that are missing.

The daily routine (~60–90 min)

When	Action	Where
Overnight	Nightly chain runs by itself: sweep → enrichment → mandate sweep → reply sync	run_sweep.bat via Task Scheduler
Morning 1	Check the four working numbers; run any engine the scheduler missed	Home
Morning 2	Work every follow-up due (pre-drafted). Most replies come from follow-ups.	Outreach Queue
Morning 3	New sends, best first, up to the daily cap (~15). Use Focus mode.	Outreach Queue
Morning 4	Work the prospect queue the same way (our own clients)	Prospects
All day	Answer every reply within 2 hours (US/EU business hours)	Inbox / LinkedIn
Before stopping	“Replies to handle” must be zero	Home

Weekly

- Run Discovery (--hn 3) to grow the watchlist — more boards = more signals = better mini-lists.
- Send at least one mini-list to a prospect agency (the strongest $0 sales asset).
- Export and send each client their CSV report from their Client Page.

Golden rule: log every touch in the app (Sent / Followed up / They replied). The follow-up timers and the double-DM guard only work if the app knows what you did.

// PART 3

The CRM, Page by Page

■ Home — the control room

The first screen each morning: today's numbers, engine controls, and money.

- ■ **Four working numbers** — new signals today, queue size (leads still 'new'), follow-ups due, replies to handle.
- ■ **Run engines** — Sweep, Enrichment, and Discovery as buttons with live streaming output. A lockfile guarantees only one engine runs at a time; a *Clear stale engine lock* button appears if a run crashed.
- ■ **Pipeline funnel** — all leads → contacted+ → replied+ → won. Reply-rate target ≥8%; hooked messages benchmark 15–25%.
- ■ **Billable** — per client: won meetings × meeting rate.
- ■ **Setup checklist** — the four first-run requirements with exact fixes.

■ Outreach Queue — the daily assembly line

Everything ready to send today, in the right order. Follow-ups first.

- ■ **Daily cap** — max new sends today (default 15; LinkedIn-safe).
- ■ **Focus mode** — one card at a time, 'send X of Y' progress, auto-advance on any action. **Skip** → cycles a card without recording.
- ■ **Cards** — contact + LinkedIn link, job-posting proof link, and the message: new leads get the client's DM template; follow-ups get follow-up #1 or the #2 breakup. After the breakup, the lead leaves the queue forever.
- ■ **Actions** — ■ Sent (starts the timer) · ■ Followed up (resets it) · ■ They replied · ■ Dismiss · ■ Undo (restores the previous state).
- ■ **Double-DM guard** — warns if this LinkedIn contact was already contacted for another client. Don't send.
- ■ ■ **Add contact manually** — name/title/LinkedIn saved per COMPANY into the enrichment cache: one save covers every lead of that company.

■ Daily Inbox — the full pipeline

Every signal ever captured, newest first, 25 per page, filterable by client and stage.

- ■ Colored stage badge per row; 'follow-up due' is a virtual filter (contacted + timer expired); ■ marks leads with notes.
- ■ Inside a row: signal facts, contacts, free-text **Notes** (searchable from the client page), editable draft DM with copy box, one-click **Compose email** when a real address exists.
- ■ Stage dropdown + update, follow-up reset, and the same manual-contact popover as the Queue.

■ Prospects — OUR pipeline

Recruitment agencies we want as clients. Own files, own stages (new → contacted → replied → call_booked → client_won), same timers.

- ■ **Three engine buttons** — Prospector (Apollo search for founders/MDs of 2–20-person agencies; run `--probe` once first), Agency board discovery (finds prospects' own job boards), Mandate sweep (their new postings = our freshest hooks).
- ■ **Cards** — person + LinkedIn, best available hook (fresh mandate > niche fallback > 'no hook — don't send'), DM from prospect templates.
- ■ **Niche keywords** — auto-derived from the agency's mandates; editable and persisted per prospect.

- ■ **One-click mini-list** — builds “10 companies hiring exactly what you place” from our tracked jobs, CSV + branded HTML, right on the card. Attach it, then: “want 50 more per week?”

■ Client Pages — per-client reporting

- Header: label, active/paused, meeting rate, keywords.
- Filters: stage / free search (covers your notes too) / period.
- Metrics: Leads, Contacted+, Replied, Won, **Billable**.
- ■ **Export CSV** — the filtered table, ready to send as the weekly client report.
- ■ **Email tool** — Instantly-format export of leads with unlocked emails + **Sync replies** from a configured adapter (Part 7).

■ Mini-List — the door-opener

A branded gift list of companies hiring in a PROSPECT agency's niche, built from boards we already track.

- Inputs: prospect name (goes in the headline), THEIR niche keywords and market, list size (10 is the sweet spot), footer pitch line.
- Outputs: raw CSV + branded one-pager HTML — open, Ctrl+P → PDF, attach.
- Companies with a named contact and multiple matching roles rank first.

■ Live Jobs — the raw pool

Every live job currently tracked — what mini-lists, dry-runs and backfills draw from.

- Substring search, platform multiselect, location filter, CSV export.

■ Watchlist — the inventory

The boards swept nightly. More boards = more signals.

- **Paste any board URL** — platform + token auto-detected.
- **Bulk import** — paste anything (page source, HN thread); every board URL is detected and added in one click.
- ■ **Discovery engine** — harvest HN threads from inside the app.
- Search across boards, live-job count per board, bulk-remove.

■■ Client Configs — the routing brain

One JSON per client. The sweep routes every new job through every ACTIVE profile.

- **Gates** — exclusion keywords (title must contain none) → inclusion keywords (≥1; each match = 10 pts) → allowed locations ('remote' also matches remote-flagged titles).
- **Outreach & billing** — DM / email / follow-up #1 / breakup templates (placeholders: {first_name} {hook} {company} {title} {location} {client}), meeting rate, delivery channel + campaign id (Part 7).
- **Test before you trust** — live single-job tester with DM preview, and a dry run over every tracked job.
- ■ **Backfill** — seed a new client with today's live matches instead of waiting for future diffs. Duplicates skipped.
- ■ **Delete** — requires typing the exact client ID.

// PART 4

The Engines (command-line reference)

Engines are standalone scripts: each reads/writes specific files and knows nothing about the app. Run them from the app folder. Most are also buttons in the app (Home and Prospects pages).

Engine	What it does	Key commands
signal_engine.py	Nightly sweep: fetch all boards, diff, emit signals with hooks	python signal_engine.py --baseline (very first run only)
discovery_engine.py	Grow the watchlist: HN 'Who is hiring' harvest, name-list probing, page extraction; dead-candidate cache (90 d)	--hn 3 [--keyword java] --names companies.txt --extract page.html · --dry-run
enrichment_engine.py	Apollo lookup of 2 decision-makers per signal company; cached forever per company	--probe gitlab (verify first!) --limit 20
prospector_engine.py	Apollo search for founders/MDs of small recruitment agencies → prospects.csv (dedup'd, never overwrites your edits)	--probe (ALWAYS before first real run) then plain runs
agency_discovery.py	Classify prospects' boards as agency boards → agency_watchlist.json (never mixed into the client watchlist)	--from-prospects
prospect_signal_engine.py	Mandate sweep over agency boards → prospect_signals.csv ('they just took a mandate' hooks)	python prospect_signal_engine.py
minilist_engine.py	Branded gift list for a prospect agency from tracked jobs	--prospect "Acme" --keywords "java, .net" [--locations …]
delivery.py	Email-tool layer: adapters for Instantly / Smartlead / Apollo sequences; lead push, CSV export, reply sync. No-ops without keys.	--probe (print would-be requests) --sync-replies

Probe-first rule: any engine that talks to a paid API has a --probe flag. Run it once with your real key before trusting a real run — vendors change payloads, and probes catch that for free.

// PART 5

Business Rules — the logic that protects you

Rule	How it works	Why it exists
Follow-up timers	A contacted lead comes due 3 days after the last touch. Follow-up #1 nudges; follow-up #2 is the breakup. After #2 the lead leaves the queue forever.	Most replies come from follow-ups; the cap stops you chasing the dead.
Hook freshness	Hooks older than 7 days automatically degrade from 'just opened' to 'currently hiring' wording.	A DM must never claim recency the posting doesn't have.
Double-DM guard	A LinkedIn contact already contacted for one client triggers a warning on every other client's card.	One person pitched twice with two offers burns both.
Baseline semantics	The first sweep ever — and the first sweep of any newly added board — records state silently. Only later diffs become signals.	No false flood of 'new' jobs that are actually old.
Daily cap	Default 15 new sends/day; ~100–150 connection requests/week.	LinkedIn account survival. The account IS the business.
Engine lock	One engine at a time, enforced by a lockfile across windows.	Two sweeps racing would corrupt the diff state.
Pipeline separation	Leads and prospects never share files, stages, or status keys. Agency boards never enter the client watchlist.	Mixing them corrupts both books.
Humans send	The app drafts everything; a human sends every LinkedIn message. No scraping, no automation on LinkedIn — ever.	ToS. Automation = account death = business death.

// PART 6

Data Files — who writes what

Everything lives as flat files in the app folder. Formats are frozen — scripts depend on the exact shapes. watchlist.json is git-tracked (curated state); the rest of the runtime data is not.

File	Contents	Written by
watchlist.json	Client-pipeline boards per platform	Watchlist page, discovery engine
signals_state.json	Last snapshot of every board = the live-jobs pool	signal engine
signals.csv	Append-only signal log (date, client, company, title, …, hook)	signal engine, backfill
inbox_status.json	Lead stages, dates, notes, follow-up counts	the CRM (via stage helpers)
enriched_contacts.json	company → contacts cache; companies never re-queried	enrichment engine, manual contact form
configs/*.json	One client profile per file	Client Configs page
discovery_state.json	Dead discovery candidates, 90-day expiry	discovery engine
prospects.csv	Our prospect agencies + people (your edits are preserved)	prospector engine + YOU
prospects_seen.json	Prospector dedupe cache	prospector engine
agency_watchlist.json	Agency boards (never merged into watchlist.json)	agency discovery
prospect_signals.csv / _state.json	Mandate signals + sweep state	prospect signal engine
prospects_status.json	Prospect stages/notes/niche overrides	Prospects page
prospects_config.json	Prospect DM/follow-up/breakup templates	Prospects page
.env	API keys (never committed)	you

// PART 7

Automation & Integrations

7.1 Scheduled automation (free, do this now)

Two batch files chain the engines; Windows Task Scheduler runs them. Steps that need missing API keys simply log a line and move on, so the chain is safe to schedule before any tool is paid for.

run_sweep.bat — nightly (e.g. 07:00):

```
cd /d C:\Users\harJD\Downloads\AgenticAILeads\app
python signal_engine.py (new signals)
python enrichment_engine.py (contacts — needs Apollo key)
python prospect_signal_engine.py (agency mandate hooks)
python delivery.py --sync-replies (email replies -> 'replied')
```

run_weekly.bat — weekly (e.g. Sunday 09:00):

```
python discovery_engine.py --hn 1 (fresh HN thread -> watchlist)
python agency_discovery.py --from-prospects (new agency boards)
```

Register both (run once in Command Prompt):

```
schtasks /Create /TN "AAL Nightly" /SC DAILY /ST 07:00 ^
/TR "C:\Users\harJD\Downloads\AgenticAILeads\app\run_sweep.bat"
schtasks /Create /TN "AAL Weekly" /SC WEEKLY /D SUN /ST 09:00 ^
/TR "C:\Users\harJD\Downloads\AgenticAILeads\app\run_weekly.bat"
```

Everything logs to `sweep.log` in the app folder — check it if a morning looks empty.

7.2 Apollo — trial first, then paid

- **The 14-day Professional free trial** unlocks the API before any money is spent. Start it when you can use two focused weeks: run `enrichment_engine.py --probe` and `prospector_engine.py --probe` first, then harvest hard: enrich every signal company and run the prospector daily. Everything lands in local caches (enriched_contacts.json, prospects.csv) that remain yours after the trial ends.
- **After the trial:** if a client has signed, pay Professional (~$79/user/mo, one seat — the master API key is shared by all scripts). If not, fall back to the manual contact form and manual sourcing until revenue arrives; nothing breaks.
- **Plug-in is one line:** APOLLO_API_KEY=… in .env. All engines pick it up automatically. People-search API calls cost zero credits.

7.3 Email tools — the delivery layer

Outreach is LinkedIn-manual by default. Email automation plugs in with one key in .env — no code changes. Nothing ever sends without a configured key AND a button press.

Step	Action
1	Warm infrastructure first: 2–3 lookalike domains (~$12/yr) + 2 inboxes ($7/mo each), SPF/DKIM/DMARC set, 4–6 weeks of warmup ramping 5–10 emails/day. Never send volume from the main domain.
2	Choose the tool: Apollo Sequences (included in Professional — start here), Instantly ($37/mo) or Smartlead when scaling. Add its key to .env.

Step	Action
3	Verify payloads: python delivery.py --probe prints the exact requests each adapter would send. Confirm any TODO(verify) fields once.
4	Per client (Client Configs): set channel to email or both, paste the tool's campaign id.
5	Push leads from the client page (only leads with real unlocked emails go). Personalization variables {hook} and {title} travel with each lead.
6	Replies sync back nightly (run_sweep.bat) or via the Sync replies button — matched by email, stage flips to 'replied', follow-ups stop automatically.

Buying order with the first ~$300–500: 1) Apollo Professional ($79) — data + sequences in one. 2) Domains + inboxes (~$40 + $14/mo) — start the warmup clock the same day. 3) Instantly ($37/mo) — only when scaling inboxes. 4) Clay ($149/mo) — only after client #2–3.

// PART 8

Worked Example — a mock client, end to end

Meet Northbridge Technical Recruiting (fictional): an 8-person agency in Manchester, UK. They place Java and DevOps contractors with UK fintechs. Founder: Sarah Whitfield. Here is the complete lifecycle, exactly as the app handles it.

Act 1 — Northbridge becomes our prospect

Day	What happens	Tool
0	The prospector (or a LinkedIn search) surfaces Sarah — founder, 2–20-person staffing agency, UK, active poster. She lands in prospects.csv.	prospector_engine / manual
0	Agency discovery probes 'northbridge' and finds their Workable board: 11 live postings — their client mandates.	agency_discovery.py
1	The mandate sweep baselines the board silently.	prospect_signal_engine
3	New posting detected: 'Senior Java Developer — Leeds, hybrid'. Prospect signal created with hook.	prospect_signal_engine

On the Prospects page, Sarah's card now shows a fresh hook and auto-derived niche keywords (java, devops, kubernetes). We click the one-click mini-list — 10 UK companies currently hiring Java/DevOps, with contacts — download the branded one-pager, and send this DM from the co-founder's LinkedIn:

> “Hi Sarah — saw you're working a Senior Java mandate in Leeds. That's exactly the market we watch: attached are 10 UK companies hiring Java/DevOps right now, with the people to talk to — yours free. We book direct-client meetings in your niche and you pay only per meeting that happens. First one's free. Want 50 of these a week?”

Click ■ **Sent**. Day 3, no reply → the queue surfaces follow-up #1 pre-drafted. Sarah replies day 4 → ■ **They replied** → call booked → stage **call_booked**. On the call (agenda in PLAYBOOK.md): what's a good client, what roles, what's a mandate worth, who decides. Close: “I build the list, you approve, I start conversations. Pay per showed meeting, first free.” Stage: **client_won**.

Act 2 — Northbridge becomes a configured client

Client Configs → new profile, five minutes:

```
{
"client_id": "northbridge",
"label": "Northbridge — UK Java/DevOps contractors",
"active": true,
"inclusion_keywords": ["java", "devops", "kubernetes", "spring"],
"exclusion_keywords": ["intern", "graduate", "unpaid"],
"allowed_locations": ["united kingdom", "london", "manchester",
"leeds", "remote"],
"keyword_weights": {"contract": 5, "senior": 2},
"meeting_rate": 150,
"dm_template": "Hi {first_name} — {hook}. Contract Java/DevOps is
exactly what we place across UK fintech. Worth a quick chat about
filling it faster?"
}
```

The live tester passes a sample job; the dry run shows **34 matches** among currently tracked jobs. Click ■ **Backfill** — Northbridge starts with 34 live leads today instead of waiting for tomorrow's diffs. We also run `discovery_engine.py --hn 3 --keyword java` to stock more UK-relevant boards.

Act 3 — the daily loop earns the money

Day	Event	In the app
1	Overnight sweep: 'Fintova' (watched fintech) opens 'Senior Java Engineer — Remote, UK'. Signal routed to northbridge, score 22.	appears in Queue
1	Contact: James Okafor, Head of Talent (enrichment cache). DM drafted: 'Hi James — saw you just opened a Senior Java Engineer role (remote, UK). Contract Java is exactly what we place…' Sent on LinkedIn → ■	Queue card → contacted
4	No reply → follow-up #1 pre-drafted, sent, ■ logged.	Queue (follow-ups due)
5	James replies 'open to a chat'. ■ → stage replied. We introduce James to Sarah; intro call scheduled.	Inbox, notes logged
8	Meeting happens — James showed. Stage → won. Billable: $150.	Client page metrics
30	Month one: 41 leads worked, 6 meetings showed → $900 invoiced. Weekly CSV reports sent from the client page.	■ Export CSV

The same loop, fully automated later: once domains are warm and a key is in .env, Fintova-type leads with unlocked emails get pushed to an email sequence instead (channel: both). The sequence sends and follows up by itself; replies sync back and flip stages. The humans only handle conversations and calls.

// PART 9

Troubleshooting

Symptom	Cause & fix
No signals after a sweep	First run only records the baseline — signals are differences between runs. Run again tomorrow. Also check: watchlist non-empty, at least one config active.
Apollo 403	Scoped key → enable master API key (Apollo → Settings → Integrations → API), update .env. If the message says the endpoint isn't on your plan: the free plan gates it — use the trial/paid plan or the manual contact form.
Board 'not found (check token)'	Wrong token or the company changed ATS. Open the board URL in a browser; remove dead boards; re-find via site:boards.greenhouse.io .
A lead isn't in the queue	Most likely: dismissed, breakup already sent (by design), below today's cap, or filtered out — the config dry-run shows exactly what's caught.
'Engine lock present' but nothing runs	A previous run crashed. Click 'Clear stale engine lock' in the engine panel.
Empty morning / no overnight data	Check sweep.log in the app folder — every scheduled step logs there. Verify the Task Scheduler tasks exist and the .bat path is correct.
Email push button disabled	No adapter configured — add the tool's key to .env and set channel + campaign id in the client's config (Part 7.3).
Windows git push fails (TLS)	The machine's TLS-inspecting AV breaks git's CA store — the repo sets http.sslBackend schannel locally; use git config --global http.sslBackend schannel for clones outside the repo.

Operating rules that keep the business alive

- Sends before systems — outreach hours are sacred; build in the gaps.
- Every reply answered within 2 hours (US/EU business hours).
- One niche until $5k/mo. Deliverability is the business. No fake proof, ever.

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